Effects of High-frequency Operation on Magnetic Components in Power Converters

Jun Imaoka
Institute of Materials and Systems for Sustainability (IMaSS)
Nagoya University
Nagoya, Japan
imaoka@nuee.nagoya-u.ac.jp

Wu Yu-Hsin
Department of Electrical Engineering
Nagoya University
Nagoya, Japan
wu.yu-hsin@f.mbox.nagoya-u.ac.jp

Koichi Shigematsu
Institute of Materials and Systems for Sustainability (IMaSS)
Nagoya University
Nagoya, Japan
shige@imass.nagoya-u.ac.jp

Tatsuya Aoki
Department of Electrical Engineering
Nagoya University
Nagoya, Japan
aoki.tatsuya@k.mbox.nagoya-u.ac.jp

Mostafa Noah
Institute of Materials and Systems for Sustainability (IMaSS)
Nagoya University
Nagoya, Japan
mostafa.noah@ieee.org

Masayoshi Yamamoto
Institute of Materials and Systems for Sustainability (IMaSS)
Nagoya University
Nagoya, Japan
m.yamamoto@imass.nagoya-u.ac.jp

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***Abstract*— Compound semiconductor devices such as Gallium Nitride (GaN) and Silicon Carbide (SiC) are actively adopted in power converters to improve power conversion efficiency, downsize passive components and miniaturize cooling systems. However, in the power converters, magnetic components often become bottlenecks in power converters from size, power losses, and temperature rise perspective. Therefore, in some cases, the attractive performance of GaN and SiC cannot be fully utilized due to the limits imposed by magnetic components. To indicate this matter from an experimental perspective, this paper conducts a fundamental investigation of the downsizing effect of magnetic components and temperature rise when varying switching frequency (85 kHz, 150 kHz, 350 kHz, 1 MHz) of a 1 kW boost converter as a case study. As results of experiments, although high-switching frequency operation contributes to downsizing magnetic component effectively, the thermal runaway of the magnetic core while driving at high frequency is confirmed due to downsized small surface area and increasing iron loss due to temperature dependency of Mn-Zn ferrite magnetic cores. For a fair comparison, the area product approach is used in the design procedure for selecting core size. The effect of the high-frequency drive is discussed from theoretical and experimental viewpoints.**

***Keywords*— high-frequency drive, iron loss, modeling, area product, temperature dependency, downsizing effect.**

I. Introduction

Recently, the importance of high-power density and high-efficiency power conversion is increasing in various industries such as electrified mobility and data centers for information and communication technology [1]-[5]. In general, power converters consist of power devices, their cooling system, passive components such as inductors, capacitors, transformers, and control circuits. However, passive components and cooling systems are generally bulky and they occupy significant space in power converters [1]-[4]. Therefore, to downsize passive components, applying advanced circuit topologies including interleaved, multilevel, and multi-cell are some of the attractive options [3]-[5]. In addition, advanced magnetic design such as integrated magnetic components used in advanced circuit topologies [6]-[7], and active buffer technologies for downsizing electrolytic capacitors [8] are also respectively attractive options. On the other hand, the selection of appropriate passive components considering power level, frequency, temperature considering ambient and self-heating, and power loss characteristics are important matters.

Within these downsizing ways from the history of power electronics growth, a high-frequency drive of power devices is a simple way for downsizing passive components because the desired capacitance and inductance decided by circuit specification can be reduced. However, it is often difficult when using Si-based power devices that have high switching and conduction losses. On the other hand, these problems can be solved by applying compound semiconductor devices such as GaN and SiC with attractive features such as high switching speed, low on-resistance, and low-gate charge characteristics. However, to effectively use compound semiconductor devices, magnetic material selection of magnetic components is a strongly important issue [9].

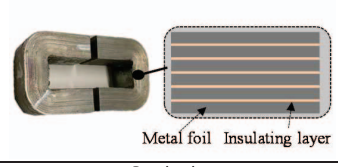
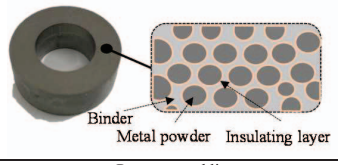
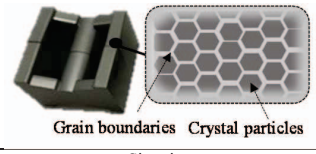
A. A simple review of magnetic materials used in the power converter and their pros and cons

First of all, features of several magnetic materials used in power converter are reviewed by dividing the classification of the laminated core, powder core, and ferrite cores. Their performances were effectively reported in [10]-[14]. Table I shows the basic magnetic core structures and their pros and cons.

1) Laminated cores: laminated cores are made by consist of metal foils and insulating layers, and these are laminated. The reason for the laminated core structure is to reduce eddy current losses, and the thickness of the metal foil of the commercial cores is dozens of μm to hundreds of μm. As the attractive feature of laminated core is high-relative permeability and high-saturation flux density. Therefore, when using a transformer in an isolated dc/dc converter with 100 kW class, non-cut Nanocrystalline cores with low-core losses are contributing to downsizing and improving efficiency as reported in [15]. However, when using as a gapped inductor using a laminated core, the eddy current losses at surfaces of gap edges due to the fringing flux increases. To mitigate the eddy current loss at gap edge due to fringing flux, split-core structure [16] and optimizing the shape of the air gap [17] have been respectively proposed.
2) Powder cores: powder cores are made by metal or alloy powders and binders while applying pressure

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TABLE I
BASIC MAGNETIC MATERIAL PROPERTIES

	Laminated cores	Powder cores	Ferrites
Magnetic core			
Processing	Lamination	Pressure molding	Sintering
Advantage	High saturation flux density High Curie temperature High relative permeability	High saturation flux density High Curie temperature Soft saturation	Low iron loss High relative permeability
Disadvantage	Increasing core loss at high frequency range	Low permeability	Low thermal immunity (low Curie temperature) Low saturation flux density
Application example	Low frequency transformer for high power EMI filter, and so on	Dc current superimposed inductor, PFC converter, and so on	High frequency transformer, inductor with airgap EMI noise filter, and so on

molding using a mold. Due to this magnetic structure, the distributed airgaps are constructed between metal powders. Therefore, although the relative permeability μ_r has low values (μ_r<200), powder cores are often used as inductors in high current applications such as bidirectional dc/dc converter in hybrid electric vehicles because of high saturation flux density, high Curie temperature characteristics, and soft saturation characteristics [1], [18]-[20].

3) Ferrite cores: In general, two types of ferrite core are used in power converter: Mn-Zn ferrite and Ni-Zn ferrite. Usually, Mn-Zn ferrite cores are used as magnetic cores for inductors and transformers within dozens of kHz to a few MHz range because of their low iron loss characteristics. However, a real part of the imaginary relative permeability of Mn-Zn ferrite core used in the hundreds of kHz range generally decreases from around 1MHz. Therefore, nowadays, Mn-Zn ferrite cores that can adapt to high-frequency operation up to a few MHz are being developed [21]-[23]. On the other hand, an attractive feature of Ni-Zn ferrite can wound windings to the core directly due to their high volume resistivity characteristics. Ni-Zn ferrite can operate at a more high frequency over the MHz range than Mn-Zn ferrite from a relative permeability perspective [13]. However, Ferrite cores are more likely to saturate at higher temperatures, and inductance values dramatically decrease [24]

B. The Purpose of This Research

Although magnetic material performance depends on the metal composition and structure, the aforementioned magnetic materials have respectively pros and cons. However, in the magnetic materials, Mn-Zn ferrite cores are one of the effective magnetic material candidates compatible with the high-frequency drive because iron loss properties are lower than that of laminated cores and relative permeability is higher than that of powder cores. However, when driving power devices such as GaN-HEMT, SiC MOSFET at high-frequency, there is almost no literature study discussing the impact of high-frequency operation and its harmful effects. Therefore, this paper carried out a straightforward investigation of the downsizing effect and temperature rise of magnetic components used in power converter by applying a high-switching frequency drive of power devices. The evaluation is done in a boost converter with Mn-Zn ferrite core inductors, as shown in Fig. 1. In this evaluation, the circuit

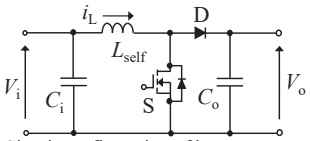


Fig. 1. Circuit configuration of boost converter.

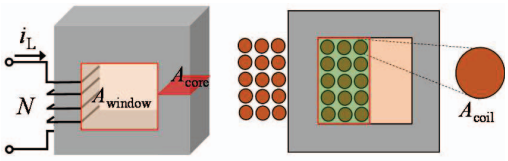


Fig. 2. Definition of winding and core geometric parameters.

parameters such as input voltage, output voltage, and inductor ripple current have the same condition. Only the switching frequency on circuit parameters is the variable parameter, it varies between 85 kHz to 1 MHz. Four inductor prototypes are designed in each switching frequency based on the area product approach [25]. The advantage and concerns of the high-frequency drive of power devices are discussed from theoretical and experimental viewpoints.

II. DESIGN OF MAGNETIC COMPONENTS

Area product (A_p) approach based on the product of the window area A_{window} and the core sectional area A_{core} as shown in Fig. 2 is one of the core size selection methods [25]. The A_p value has two meanings. One is the A_p value determined from the physical core size. Therefore, when selecting a magnetic core size, A_p value from the physical core size is determined. On the other hand, from a theoretical perspective, A_p value is calculated by the current density J_{coil} of winding and maximum flux density B_{max} of magnetic cores, and inductance L_{self} while considering circuit specification. Theoretical A_p values in the case of inductors can be calculated by the following:

$$A_p = \frac{L_{self} \cdot I_{Lmax}^2}{K_w \cdot J_{coil} \cdot B_{max}} \quad (1)$$

where K_w is the window utilization factor which is the ratio of total coil sectional area decided by the number of turns N, winding sectional area A_{coil} to window area A_{window}. Therefore, based on area product approach, inductor core size can be selected.

Then, as a circuit specification of the boost converter, Continues Current Mode (CCM) is assumed. Therefore, the maximum current I_{Lmax} of inductors, are shown as follows:

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TABLE II
CIRCIT PARAMETERS OF BOOST CONVERTER

Input voltage	V_i	100 V
Output voltage	V_o	200 V
Input power	P_i	1 kW
Switching frequency	f_s	85 kHz,150 kHz, 350 kHz,1 MHz
Duty cycle	d	0.5
Inductor average current	I_{Lave}	10 A
Inductor current ripple	I_{Lpp}	15 A
Maximum Flux density	B_{max}	0.2 T
Current density	J_{coil}	5 A/mm^2

$$I_{Lmax} = I_{Lave} + \frac{I_{Lpp}}{2} \tag{2}$$

where I_{Lave} is the inductor average current, and I_{Lpp} is the inductor ripple current. The relationship between I_{Lpp} and L_{self} is as follows:

$$L_{self} = \frac{V_i \cdot d \cdot T_s}{I_{Lpp}} = \frac{V_i}{I_{Lpp} \cdot f_s} \cdot \left(1 - \frac{V_i}{V_o}\right) \tag{3}$$

where d is the duty cycle of main switch S, T_s is the switching period, f_s is the switching frequency. Therefore, substituting (2) and (3) into (1), the following equation can yield:

$$\begin{aligned} A_p &= \frac{L_{self} \cdot I_{Lmax}^2}{K_w \cdot J_{coil} \cdot B_{max}} \\ &= \frac{\frac{V_i}{I_{Lpp} \cdot f_s} \cdot \left(1 - \frac{V_i}{V_o}\right) \cdot \left(I_{Lave} + \frac{I_{Lpp}}{2}\right)^2}{K_w \cdot J_{coil} \cdot B_{max}} \end{aligned} \tag{4}$$

Commercially available magnetic cores are usually available in the market as a series with different core sizes. For a fair comparison, we selected the same structure in all cases. In this case, EI core structures are selected in all prototypes to eliminate core structural dependency. The following core sizes EI60, EI50, EI40, and EI30, are selected respectively. The magnetic material is Mn-Zn ferrite cores and they are commercially available from a supplier. The circuit parameters of the boost converter are as shown in Table II. By comparing A_p value from physical core size and theoretical line based on (4), core size is selected in each switching frequency condition as shown in Fig. 3. As a policy of this comparison when varying switching frequency, the ratio of the A_p values determined by the physical core size to that determined by the electromagnetic requirements based on (4) is set to 0.41 for a fair comparison.

A. *Calculating the number of turns, magnetic reluctance, and inductance*

After selecting magnetic core size in each switching frequency based on area product approach, the number of winding turns N, magnetic reluctance R_m on the closed magnetic path, and inductance L_{self} are respectively designed. Figure 4 shows the magnetic circuit model of an inductor. where Ni_L is the magnetomotive force, R_m is the magnetic reluctance including magnetic reluctance at air gap, ϕ is the magnetic flux. Based on the magnetic circuit, the following equation can be obtained:

$$\begin{cases} \phi_{max} = \dfrac{N \cdot I_{Lave}}{R_m} + \dfrac{1}{2} \cdot \dfrac{V_i}{N} \cdot d \cdot T_s \\ \phi_{max} = B_{max} A_{core} \\ I_{Lpp} = \dfrac{R_m \cdot V_i \cdot d \cdot T_s}{N^2} \end{cases} \tag{5}$$

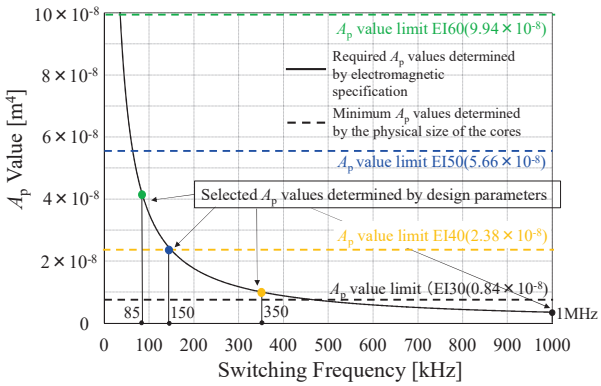


Fig. 3. A_p value comparison from physical core size, theoretical line, and design points.

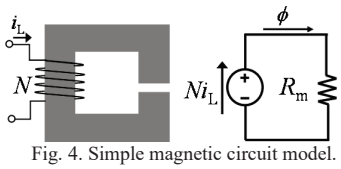


Fig. 4. Simple magnetic circuit model.

Therefore, by solving (5), the following design equations can be obtained:

$$\begin{cases} N = \dfrac{\left(2 \cdot I_{Lave} + I_{Lpp}\right) \cdot V_i \cdot d \cdot T_s}{2 \cdot B_{max} \cdot A_{core} \cdot I_{Lpp}} \\ R_m = \dfrac{I_{Lpp} \cdot N^2}{V_i \cdot d \cdot T_s} \\ L_{self} = \dfrac{N^2}{R_m} \end{cases} \tag{6}$$

By using the above equations, N, R_m, and L_{self} are determined, respectively.

B. *Design results and prototypes*

The design results are summarized in Table III and the prototypes are shown in Fig. 5. As seen in Table III, although the low-switching frequency drive (85 kHz) requires a bigger magnetic core size (core volume: 26900 mm^3), high-switching frequency as 1MHz is effective for downsizing magnetic core (in this case, core volume: 6440 mm^3). The downsizing effect of inductor core at 1 MHz drive is about 1/4 compared with that at 85 kHz drive. Besides, not only the magnetic core can be downsized but also the number of turns can be reduced and it contributes to reducing the DC resistance of the winding. This is an attractive impact of applying a high-switching frequency drive.

III. EXPERIMENTAL RESULTS

In this section, experimental tests are carried out to confirm the validity of the design methods and to highlight few concerns when driving at high-frequency. Figure 6 shows the experimental waveforms of the inductor current and drain-source voltage of the power device at different switching frequencies. The experimental waveforms are captured at the steady-state operation of the boost converter (i.e the waveform captured at 1 minute after the boost converter was started at 1 kW, continuous load condition). As seen in Fig. 6, inductors operated according to the design specifications. In this experiment, thermal evaluations are

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TABLE III
DESIGNED VALUE AND SPECIFICATION OF INDUCTOR PROTOTYPES

Magnetic material	Mn-Zn ferrite (all prototype is the same magnetic material)			
Core structure and size	EI60	EI50	EI40	EI30
Core volume	26900 mm^3	21620 mm^3	11400 mm^3	6440 mm^3
Core weight	139 g	115 g	60 g	34 g
switching frequency f_s	85 kHz	150 kHz	350 kHz	1000 kHz
Physical A_P value (decided by core size)	$9.94 \cdot 10^{-8}$ m^4	$5.66 \cdot 10^{-8}$ m^4	$2.38 \cdot 10^{-8}$ m^4	$0.84 \cdot 10^{-8}$ m^4
Designed A_P value (eq. 4) as design points	$4.14 \cdot 10^{-8}$ m^4	$2.35 \cdot 10^{-8}$ m^4	$0.99 \cdot 10^{-8}$ m^4	$0.35 \cdot 10^{-8}$ m^4
Ratio of physical A_p values to designed A_p	0.41	0.41	0.41	0.41
Inductance L_{self}	39.2 µH	22.2 µH	9.52 µH	2.5 µH
Designed Number of turns N	14 turns	9 turns	6 turns	3 turns
Magnetic reluctance R_m	5.0 A/µWb	3.6 A/µWb	3.8 A/µWb	2.7 A/µWb

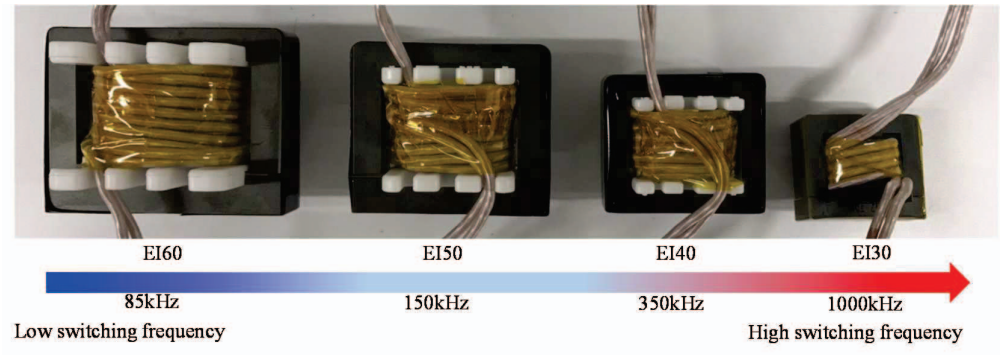
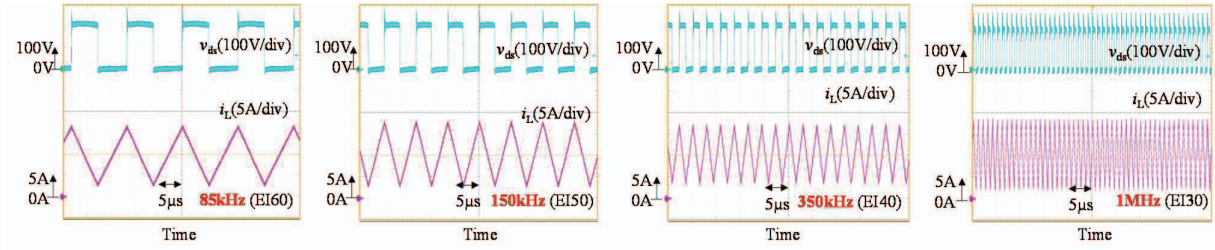
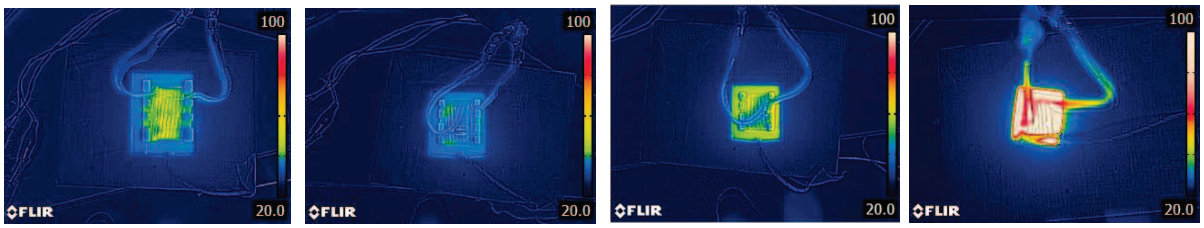


Fig. 5. Photo of the inductor prototypes and downsizing effect using high switching frequency drive.



(a) 85 kHz drive for EI60 (b) 150 kHz drive for EI50 (c) 350 kHz drive for EI40 (d) 1000 kHz drive for EI30

Fig. 6. The experimental waveforms when varying switching frequency and changing the inductor prototypes.



(a) 85 kHz drive for EI60 (b) 150 kHz drive for EI50 (c) 350 kHz drive for EI40 (d) 1000 kHz drive for EI30

Fig. 7. The temperature of each prototype captured by the thermal camera.

also carried out simultaneously using the thermal camera (FLIR E8) and a combination of data logger (GRAPHTEC midi logger GL840) and thermocouple (type K) for evaluating the temperature distribution of the inductors and for recoding thermal response in the time domain when driving boost converter, respectively. Figure 7 (a)-(d) shows the temperature distribution taken by the thermal camera. In this thermal evaluation, the measurement environment on the inductor is summarized in Fig. 8. To evaluate the temperature rise of the inductors purely, the influence of heat generation from power devices and other components is eliminated by physically enlarging the distance between components. Figure 7 (a)-(c) are the photos taken by the thermal camera when temperature rise is completely saturation condition. On the other hand, Fig. 7 (d) is the photo when using 1 MHz drive and EI 30 core, after 10 minutes from power converter is started from room temperature. As seen in Fig. 7 (d), the inductor using EI 30 has a very high-temperature rise. This is because core loss characteristics of Mn-Zn ferrites have temperature dependency as shown in Fig. 9. In general, core loss of Mn-Zn ferrites decreases along with increasing core temperature by T_p as shown in Fig. 9. However, when the core temperature reaches T_p, the thermal runaway may occur because core loss and core temperature simultaneously increase. Furthermore, it has been noted that the temperature rise also depends on the cooling method for the inductor. Regarding thermal response in the time domain measured by thermocouple and data logger are shown in section IV.

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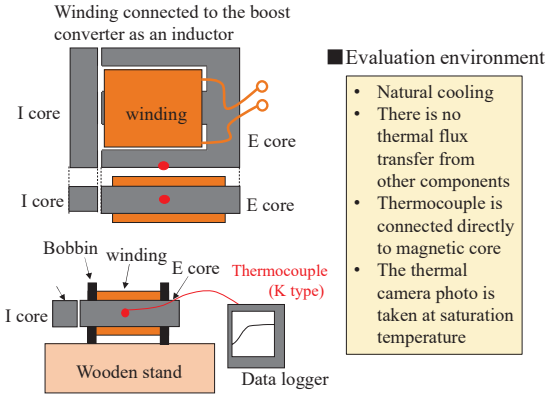


Fig. 8. Configuration of thermal evaluation.

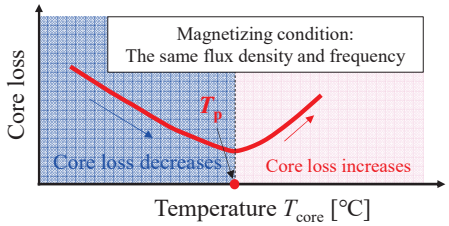


Fig. 9. Core loss characteristics of the Mn-Zn ferrite when varying core temperature.

IV. A SIMULATION MODEL BASED ON AN EQUIVALENT THERMAL CIRCUIT

In this section, analyzing and modeling the thermal response of the inductor core loss with temperature dependency for system-level simulation is conducted as a challenge. Besides, making a temperature dependent magnetic model for system-level simulation is one of the important issues to consider the temperature influence, not only ambient temperature or increasing temperature by self-heating but also thermal transfer from other components. To include the thermal response including core loss with temperature dependence, characterization of iron loss with temperature dependency is modeled using the Steinmetz equation [26].

$$P_{cv} = k \cdot f^{\alpha} \cdot B_{amp}{}^{\beta} \tag{7}$$

where P_{cv} is the core loss of the time average power loss per unit volume; B_{amp} is the peak flux density amplitude, f is the frequency of sinusoidal excitation. k, α, and β are the Steinmetz parameters and are dependent on magnetic material parameters.

However, there are limitations of applying the Steinmetz equation because the power losses are modeled by only sinusoidal excitation conditions only. Therefore, improved Generalized Steinmetz Equation (iGSE) using Steinmetz parameters k, α and β has been developed for adapting any magnetic flux waveforms focusing on dB/dt, and well known as following equation [27]:

$$\begin{cases} P_{cv} = \dfrac{1}{T_s} \cdot \displaystyle\int_0^{T_s} k_i \left|\frac{dB}{dt}\right|^{\alpha} (\Delta B)^{\beta-\alpha}\, dt \\ k_i = \dfrac{k}{(2\pi)^{\alpha-1} \cdot \displaystyle\int_0^{2\pi} |\cos\theta|^{\alpha} 2^{\beta-\alpha} d\theta} \end{cases} \tag{8}$$

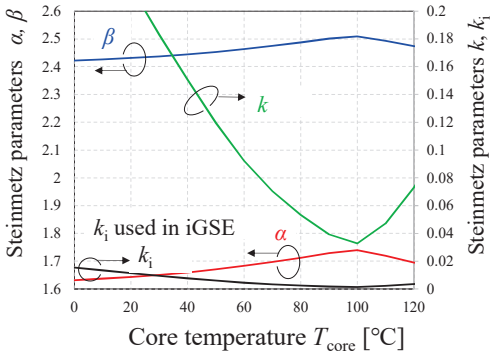


Fig. 10. Steinmetz parameters derived by datasheet from supplier.

where k_i is the coefficient used in iGSE instead of the original coefficient k of Steinmetz Equation. It is worth mentioning that this equation is not included the effect of DC biased magnetization [28]. However, utilizing (8) is an effective method because you can roughly estimate the core loss from the information available in the datasheet. Then, to consider core loss characteristics with temperature dependence, we modeled the Steinmetz parameters with temperature dependence by following four steps and using the given datasheet:

Step 1: Normalize with a value of typical temperature (e.g. T_p as shown in Fig. 9) based on the temperature characteristics of P_{cv}. This is named the Temperature Normalization Function (TNF). This core loss characteristic is a characteristic at a limited frequency and magnetic flux density in the datasheet from a manufacturer.

Step 2: For the frequency-dependent P_{cv}, find the core loss for each temperature based on the TNF defined in Step 1 with the specific magnetic flux density in the datasheet. At this time, the correction coefficients were introduced so that the P_{cv} would match the exact data points in the datasheet.

Step 3: Repeat Step 2 to supplement the magnetic flux density given to the datasheet and obtain the P_{cv} characteristics for the frequency at each temperature.

Step 4: From P_{cv} characteristics with temperature dependency, calculate k, α and β of the Steinmetz equation by using eq. (7) by curve fitting. After that, k_i is calculated based on eq. (8). Figure 10 shows the temperature-dependent characteristics k, α, β and k_i.

After parameterization of Steinmetz parameters and using iGSE, core loss is calculated by eq. (8). However, to simulate thermal response in system-level circuit simulation, a thermal circuit model is also needed. In this case, Cauer thermal circuit model is used. Figure 11 shows the Cauer circuit model used for the thermal response analysis. Where R_{core} and R_w are the thermal resistances in core and winding, respectively. $R_{c\text{-}w}$, $R_{w\text{-}a}$, $R_{c\text{-}a}$ are the thermal resistances between core and winding, between winding and ambient, between core and ambient, respectively. C_{core} and $C_{winding}$ are the thermal capacitances in core and winding, respectively. These parameters in the Cauer circuit for thermal response are evaluated from physical properties and the heat transfer coefficient and are optimized with the measured thermal response.

Figure 12 shows the simulation and experimental results for a thermal response. As seen in this figure, simulation and experimental results are matching each other by considering iron loss temperature dependence and parameterization of the

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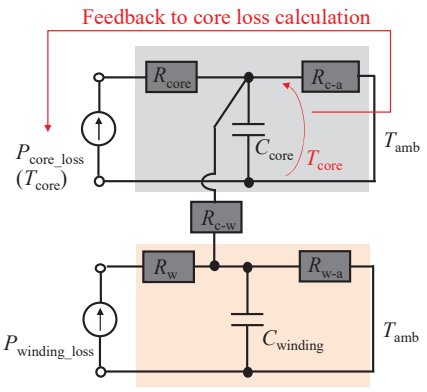


Fig. 11. Cauer thermal circuit model used in this evaluation.

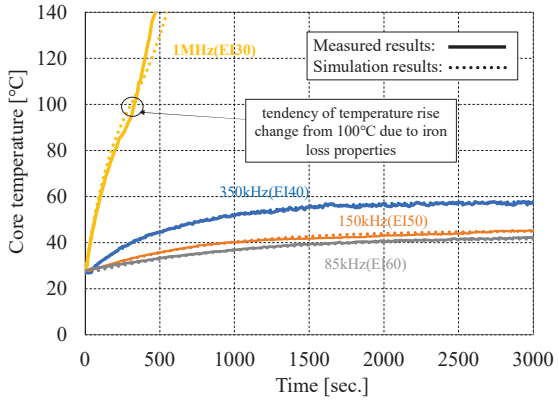


Fig. 12. Measured and simulation results of temperature rise tendency of magnetic core. The measuring point of thermocouple are shown in fig. 8.

thermal circuit model. However, the Cauer circuit model is parameterized by the thermal response from experimental results. It worth mentioning that when using this model, the increasing tendency of the core temperature can almost be estimated.

As future work, we will construct comprehensive modeling and simulation methods using thermal response for system-level simulation for front-loading design.

V. Conclusion

This paper investigated the downsizing effect and temperature rise of magnetic components used in power converter by applying a high-switching frequency drive of power devices. As a result of this evaluation, it has been concluded that driving power devices at high frequency, downsizing inductors can be realized. However, the temperature rise of magnetic components will be a critical issue to effectively use the excellent performance of compound semiconductors. Besides, simple thermal evaluations from simulation and experimental perspectives were carried out. As a result, thermal runaway due to core loss with temperature dependency was one of the important issues. Therefore, cooling of magnetic components is important for power converters. Furthermore, this paper will be one of the pieces of evidence indicating the importance of developing magnetic material for high switching frequency drive.

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